

is a significant chance that a duplicate message may arrive at the receiver. The receiver SHALL detect and mark duplicate messages that it receives using the standard authentication and replay protection mechanisms of the secure message layer (see [Section 4.6.5, “Replay Prevention and Duplicate Message Detection”](#)). The receiver SHALL send an acknowledgment message to the sender for each instance of an authenticated, reliable message, including duplicates. The reliability layer SHALL only propagate the first instance of a message to the next higher layer. Any message marked as a duplicate SHALL be dropped by the reliability layer.

4.12.3. Peer Exchange Management

The Reliable Messaging Protocol operates within the scope of an [Exchange](#) between two Nodes. MRP SHALL support one pending acknowledgement and one pending retransmission per [Exchange](#).

MRP control parameters, detailed in [Table 22, “Glossary of MRP parameters”](#), are computed outside of the [Exchange communication](#) itself; instead, they are valid for the duration of a secure session. The [SESSION_ACTIVE_INTERVAL](#) and [SESSION_IDLE_INTERVAL](#), used in computation of MRP control parameters, are determined during [Operational Discovery](#) or [Section 4.3.1, “Commissionable Node Discovery”](#). Additionally, the initiator of a secure session MAY provide these parameters in the initial [CASE Sigma1](#) or [PASE PBKDFParamRequest](#) messages, and the responder MAY provide its parameters in the corresponding protocol messages [CASE Sigma2](#) or [PBKDFParamResponse](#).

4.12.4. Transport Considerations

When the upper layer requests a reliable message over a UDP transport, the [R Flag](#) SHALL be set on that message indicating that MRP SHALL be used. Reliable messages sent over TCP, [PAFTP](#), or BTP SHALL utilize the underlying reliability mechanisms of those transports and SHOULD NOT set the [R Flag](#). Reliable messages sent over [NTL](#) SHALL utilize the underlying reliability mechanisms and SHALL not set the [R Flag](#).

4.12.5. Reliable Message Processing

4.12.5.1. Reliable Message Processing of Outgoing Messages

To prepare a given Protocol Message for transmission, the message SHALL be processed as follows:

1. Proceed to [Section 4.12.5.1.1, “Piggyback acknowledgment processing”](#).

Piggyback acknowledgment processing

1. Determine if there is a matching pending acknowledgement in the [acknowledgement table](#) for the given message by checking all of the following conditions:
 - a. If the Destination Node Id and Exchange Id of the given message and pending acknowledgement are the same
 - b. AND either
 - i. the Session Id and underlying Session Credentials of the given message and pending acknowledgement are the same